Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In the formula $\bar{x} = a + \dfrac{\sum f_i d_i}{\sum f_i}$ for finding the mean of grouped data, d_i are deviations from a of: [1]

(A) lower limits of classes (B) upper limits of classes
(C) mid-points of classes (D) frequencies of classes

Q2. If the difference between mode and median of a data is 24, then the difference between median and mean is: [1]

(A) 8 (B) 12 (C) 24 (D) 36

Q3. Consider the following frequency distribution: [1]

Class	0–5	6–11	12–17	18–23	24–29
Frequency	13	10	15	8	11

The upper limit of the median class is:

(A) 17 (B) 17.5 (C) 18 (D) 18.5

Q4. If the mean of the numbers $27 + x, 31 + x, 89 + x, 107 + x, 156 + x$ is 82, then the mean of $130 + x, 126 + x, 68 + x, 50 + x, 1 + x$ is: [1]

(A) 75 (B) 80 (C) 82 (D) 70

Q5. **Assertion (A):** If each observation of a data set is increased by 5, the arithmetic mean is also increased by 5.
Reason (R): The sum of deviations of a set of observations from their mean is always non-zero. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. For a frequency distribution, given $\sum f_i d_i = 36$, $\sum f_i = 40$, and assumed mean $a = 25$. Calculate the arithmetic mean. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the mode of the following distribution of daily wages: [3]

Wages (Rs)	100–150	150–200	200–250	250–300	300–350
Workers	4	5	12	2	2

Q8. Calculate the mean of the following data using the Step-Deviation Method: [3]

Class Interval	25–35	35–45	45–55	55–65	65–75
Frequency	6	10	8	12	4

Q9. Convert the discontinuous classes into continuous classes and calculate the mode: [3]

Class	11–20	21–30	31–40	41–50	51–60
Frequency	8	15	20	12	5

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: METEOROLOGICAL DEPARTMENT SEASONAL MONSOON RAINFALL RECORD [4 Marks]

The India Meteorological Department (IMD) observed seasonal rainfall records across various agro-climatic sub-divisions during the monsoon season to assist in agricultural crop planning and water resource management.

Rainfall (mm)	200–400	400–600	600–800	800–1000	1000–1200	1200–1400	1400–1600
Sub-divisions	2	4	6	10	8	4	2

(i) State the modal rainfall class and find its class mark. [1]

(ii) Calculate the mode of the seasonal rainfall across the sub-divisions. [2]

(iii) Find how many sub-divisions received less than 1000 mm of rainfall. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★